1) Determine whether the integral is convergent or divergent. Evaluate it if it's convergent.

a) $\int_{-\infty}^{1} a^r \, dr, \quad a>1$

$$= \lim_{t\to-\infty} \int_t^1 a^r = \lim_{t\to-\infty} \frac{a^r}{\ln(a)} \Big|_t^1 = \lim_{t\to-\infty} \frac{a}{\ln(a)} - \frac{a^t}{\ln(a)} = \boxed{\frac{a}{\ln(a)}}$$

b) $\int_{-\infty}^{0} \frac{1}{2-3x} \, dx$

$$= \lim_{t\to-\infty} \int_t^0 \frac{1}{2-3x} = \lim_{t\to-\infty} -\frac{1}{3} \ln|2-3x| \Big|_t^0 = \lim_{t\to-\infty} -\frac{1}{3}(\ln(2) - \ln|2-3t|)$$

$$= \boxed{\infty}$$

c) $\int_1^{\infty} \frac{2e^{-1/x^2}}{x^3}$

$$= \lim_{t\to\infty} \int_1^t \frac{2e^{-1/x^2}}{x^3} \, dx = \lim_{t\to\infty} e^{-1/x^2} \Big|_1^t = \lim_{t\to\infty} e^{-1/t^2} - \frac{1}{e}$$

$$= \boxed{1 - \frac{1}{e}}$$

d) $\int_0^2 \frac{1}{\sqrt{2-x}} \, dx$

$$= \lim_{t\to2^-} \int_0^t \frac{1}{\sqrt{2-x}} \, dx = \lim_{t\to2^-} -2\sqrt{2-x} \Big|_0^t = \lim_{t\to2^-} -2\sqrt{2-t} + 2\sqrt{2}$$

$$= \boxed{2\sqrt{2}}$$

2) Find the exact length of the curve

a) $x = y + y^5, \quad 1 \le y \le 2$ (don't evaluate)

$$\sqrt{1+\left(\frac{dx}{dy}\right)^2} = \sqrt{1+(5y^4+1)^2} = \sqrt{25y^8+10^4+2}$$

$$L = \int_1^2 \sqrt{25y^8+10^4+2}$$

b) $y = \sqrt[4]{(4(x+3)^3)} \qquad 0 < x \le 1$

$$\frac{dy}{dx} = \frac{1}{\sqrt{(x+3)^3}} \cdot 3(x+3)^2$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{9(x+3)^4}{(x+3)^3} = 9(x+3)$$

$$L = \int_0^1 \sqrt{1+9(x+3)} = \frac{2}{27}(9x+28)^{3/2}\Big|_0^1 = \frac{2}{27}\left(56\sqrt{7} - 37\sqrt{37}\right)$$

3) Find the exact area of the surface obtained by rotating the curve about the x-axis

$$y = \sqrt{1+e^x}, \quad 0 \le x \le 1$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1+e^x}} \cdot e^x, \quad \left(\frac{dy}{dx}\right)^2 = \frac{e^{2x}}{4+4e^x}, \quad 1+\left(\frac{dy}{dx}\right) = \frac{e^{2x}+4+4e^x}{4+4e^x} = \frac{(e^x+2)^2}{4(1+e^x)}$$

$$S = \int_0^1 2\pi y\sqrt{1+(y')}\,dx = \int_0^1 2\pi\sqrt{1+e^x}\,\frac{\sqrt{(e^x+2)^2}}{\sqrt{4(1+e^x)}} = \int_0^1 2\pi\sqrt{1+e^x}\,\frac{(e^x+2)}{2\sqrt{1+e^x}}$$

$$= \int_0^1 \pi(e^x+2)\,dx = \pi e^x + 2\pi x\Big|_0^1 = \pi e + 2\pi - \pi = \pi e + \pi$$

4) Find the resulting surface when the curve below is rotated about the x-axis

$$y = -\tfrac{1}{2}\ln(x) + \frac{x^2}{4}, \quad 1 \le x \le 2$$

$$\frac{dy}{dx} = \frac{-1}{2x} + \frac{x}{2}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{x^2}{4} + \frac{1}{2} + \frac{1}{4x^2} = \left(\frac{x}{2} + \frac{1}{2x}\right)^2$$

$$S = \int_1^2 2\pi x \sqrt{\left(\frac{x}{2} + \frac{1}{2x}\right)^2}\,dx = \int_1^2 2\pi x\left(\frac{x}{2} + \frac{1}{2x}\right)dx = \int_1^2 \pi(x^2+1)\,dx$$

$$= \pi\left[\frac{x^3}{3} + x\right]\Big|_1^2 = \pi\left(\frac{10}{3}\right)$$

5) Use the comparison test to determine whether the integrals diverge or converge.

$$\int_1^\infty \frac{2x}{\sqrt{x^7+3x+2}}\,dx, \quad \frac{2x}{\sqrt{x^7+3x+2}} \le \frac{2x}{\sqrt{x^7}} = \frac{2}{x^{5/2}}$$

Since $\int_1^\infty \frac{1}{x^p}\,dx$ converges only if $p > 1$

$\int_1^\infty \frac{2}{x^{5/2}}\,dx$ also converges b/c $p = 5/2 > 1$

Hence, $\int_1^\infty \frac{2x}{\sqrt{x^7+3x+2}}\,dx$ converges.